

Chee Ying “Ludwig” Tay

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EDUCATION

Purdue University, School of Science, School of Engineering | West Lafayette, IN
Bachelor of Engineering, Bachelor of Science
Majors: Electrical Engineering, Computer Science
Dean's List & Semester Honors Fall '24 Spring '25 Fall '25 Spring '26

Expected May 2028
GPA: 3.94 / 4.00

PROJECTS

Artificial Optical Neural Network | West Lafayette, IN | *Researcher*
- Building a multilayer, all-optical artificial neural network under the guidance of Professor Du and in collaboration with his graduate students.
- Developed hardware aware complex number training for optical neural networks. Developed novel application for illustrating optical nonlinearity

Learning Generalizable Predictors for Quantum Systems | West Lafayette, IN | *Researcher*
- Identified stochastic learning models as a gap in existing work in machine learning for quantum systems.
- Developed a world-model inspired JEPa and Diffusion model for predicting decoherence in solid-state quantum emitters. Established rigorous and benchmark for multiple machine learning models for comparison.

HybridMimic: Hybrid RL-Centroidal Control for Humanoid
Motion Mimicking | West Lafayette, IN | *Researcher (First Author) @ TRACE Labs*
- Developed a novel control algorithm combining Reinforcement Learning and Model Based Control to control a humanoid robot.
- Research allows for greater positional accuracy of trained robot policies for complex locomotion tasks.
- Presented poster at 2026 ICON seminar, 2026 Midwest Robotics Workshop. Accepted IROS 2026.

VLA for Human Robot Interaction | West Lafayette, IN | *Researcher @ TRACE Labs*
- Implemented VLA models to semantically control humanoid robots on G1 Humanoid.
- Exploring world models, VLA w/ diffusion decoder, and other imitation learning techniques for safe human robot interaction.

Novel Humanoid Robot | West Lafayette, IN | *Software Lead @ Humanoid Robotics Club*
- Designed a novel humanoid robot with a small team. Provided kinematics feedback through simulation and reinforcement learning.
- Trained a deep reinforcement learning based walking controller and integrated it with a low level PD controller in ROS2.
- Presented robot as first full scale humanoid at Open Sauce 2026.

Foundational AI Language Research | Cambridge, MA | *Researcher*
- Conducted AI language model research under Prof. Stuart Shieber at Harvard University, trained early AI language models on computational linguistic tasks to understand the limitations of LSTM neural networks.
- Explored augmented LSTM neural networks with a novel embedded stack mechanism to allow it to remember arbitrarily long sequences.

Table Cleaning Robot | Singapore, Singapore | *Software Intern*
- Developed table cleaning robot consisting of an autonomous base with a 12 DOF arm with a cleaning tool.
- Programmed ROS-based cleaning robot to autonomously navigate to a table, capture 3D surface imagery, and motion plan a cleaning route for the arm that cleans while avoiding collisions.
- Demonstrated robot at a public food court to Singapore's National Environmental Agency, contributing to Weston Robotics winning a \$1,000,000 bid for a future food court table cleaning robot.

PROFESSIONAL EXPERIENCE

Republic of Singapore Air Force | Singapore, Singapore | *Sergeant*
- Graduated from Military Police Leadership School; attained rank of Sergeant. Posted to Tengah Air Base to lead security troopers in protecting key Singaporean military installations.
- Received Marksmanship award and the highest physical fitness accreditation (Individual Physical Proficiency Test Gold award)
- Entrusted with leading a team securing an island airstrip with limited supervision. Enforced discipline and Professionalism for the men under my charge. Defused a high pressure situation professionally.

Indigo Technologies | Woburn, MA | *Software Intern*
- Developed hardware and software to test the company's novel motors. Added controller hardware for additional data acquisition.
- Wrote software to organize and visualize collected data, streamlining data collection and analysis pipelines to accelerate R&D cycles.

LEADERSHIP & ACTIVITIES

Humanoid Robotics Club | West Lafayette, IN | *Software Lead*
- Leading and training a team of software engineers to build and program a humanoid robot.
- Presented our prototypes and progress to important club sponsors to secure their support of the club.

Lexington High School Vex Robotics | Lexington, MA | *Founder*
- Founded and led a lasting competitive VEX robotics team that was student orientated with no adult coaching.
- Fundraised for the team, obtaining capital through self organized summer camps and sponsorships.
- Built and programmed internationally competitive VEX robots. Qualified for VEX World Championships in 2021 and 2022.

RELEVANT SKILLS

Git | Python | C | C++ | Java | JAX | Pytorch | Mujoco | IsaacSim | ROS2 | Reinforcement Learning | Optimal Control | VLA Models | Model Based Control | AI | SPICE | KiCAD | Diffusion Models | Latex